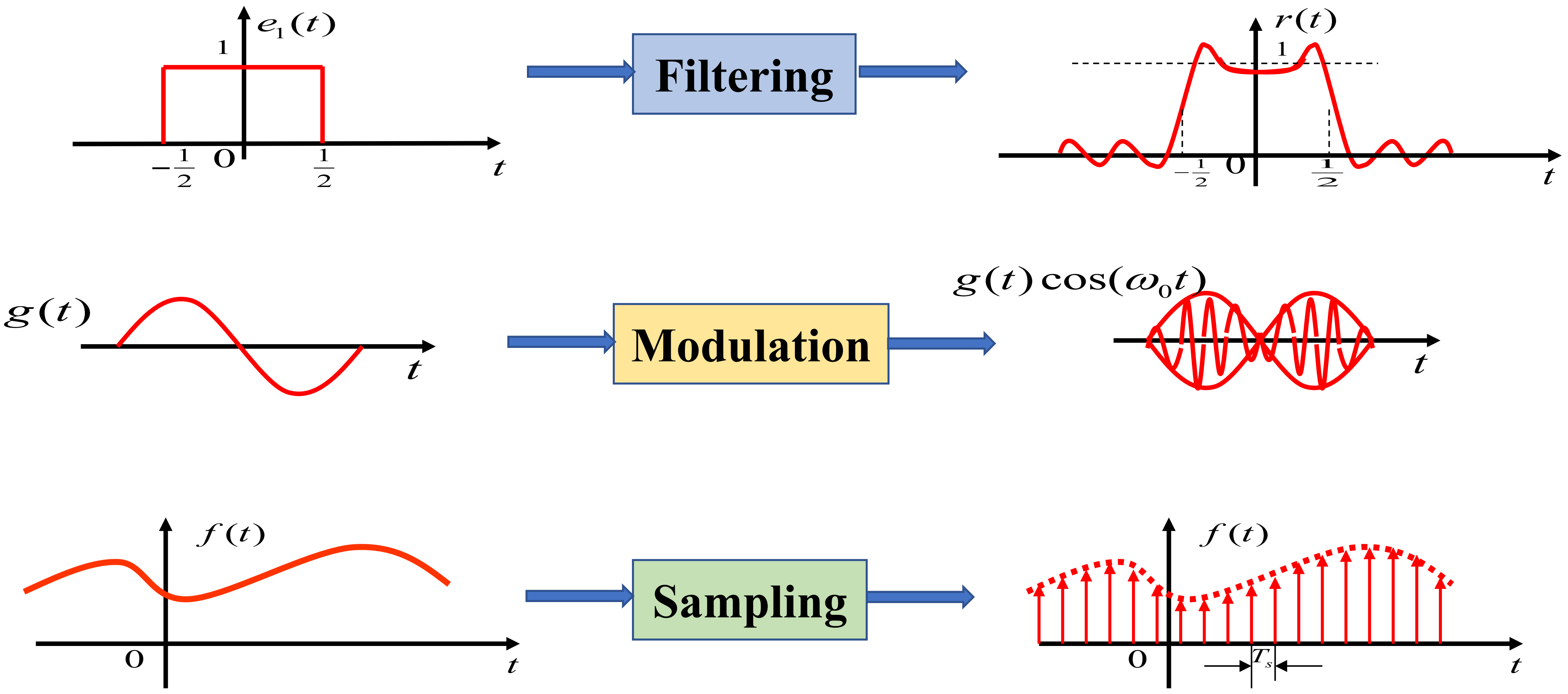
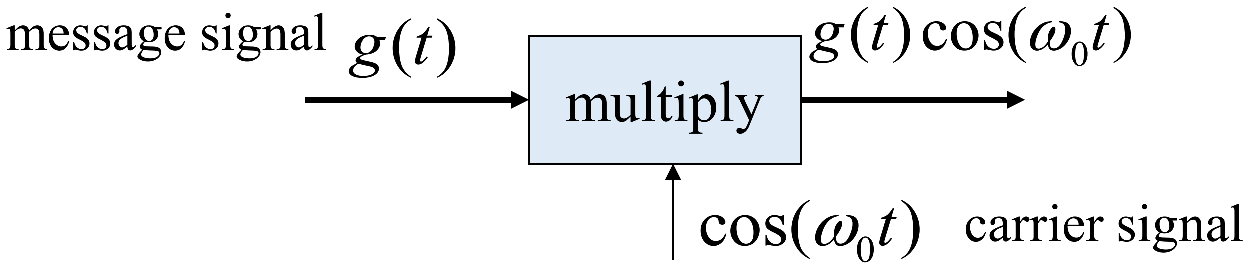
Problem Class 3

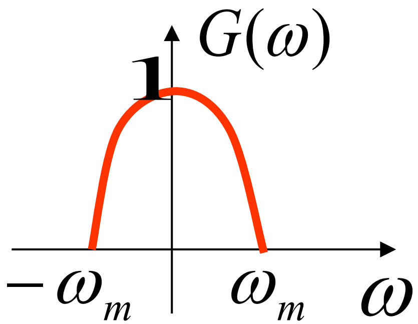
(Chapters 5~8)

Fourier transform for communication systems reveals signals' inherent frequency characteristics and the close relationship between their time and frequency characteristics, deriving key concepts such as spectrum and bandwidth.



Modulation Principle

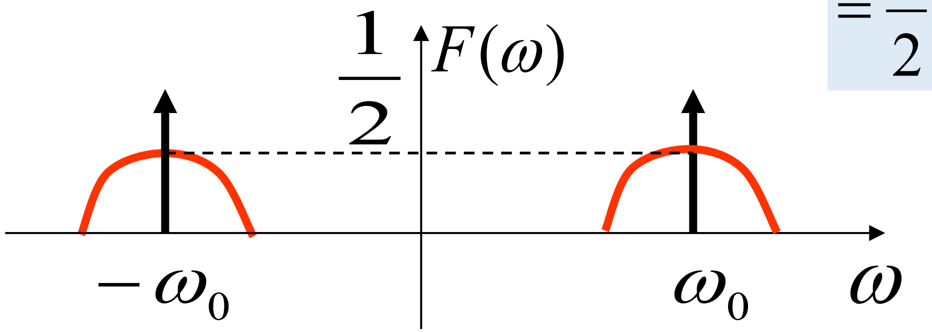


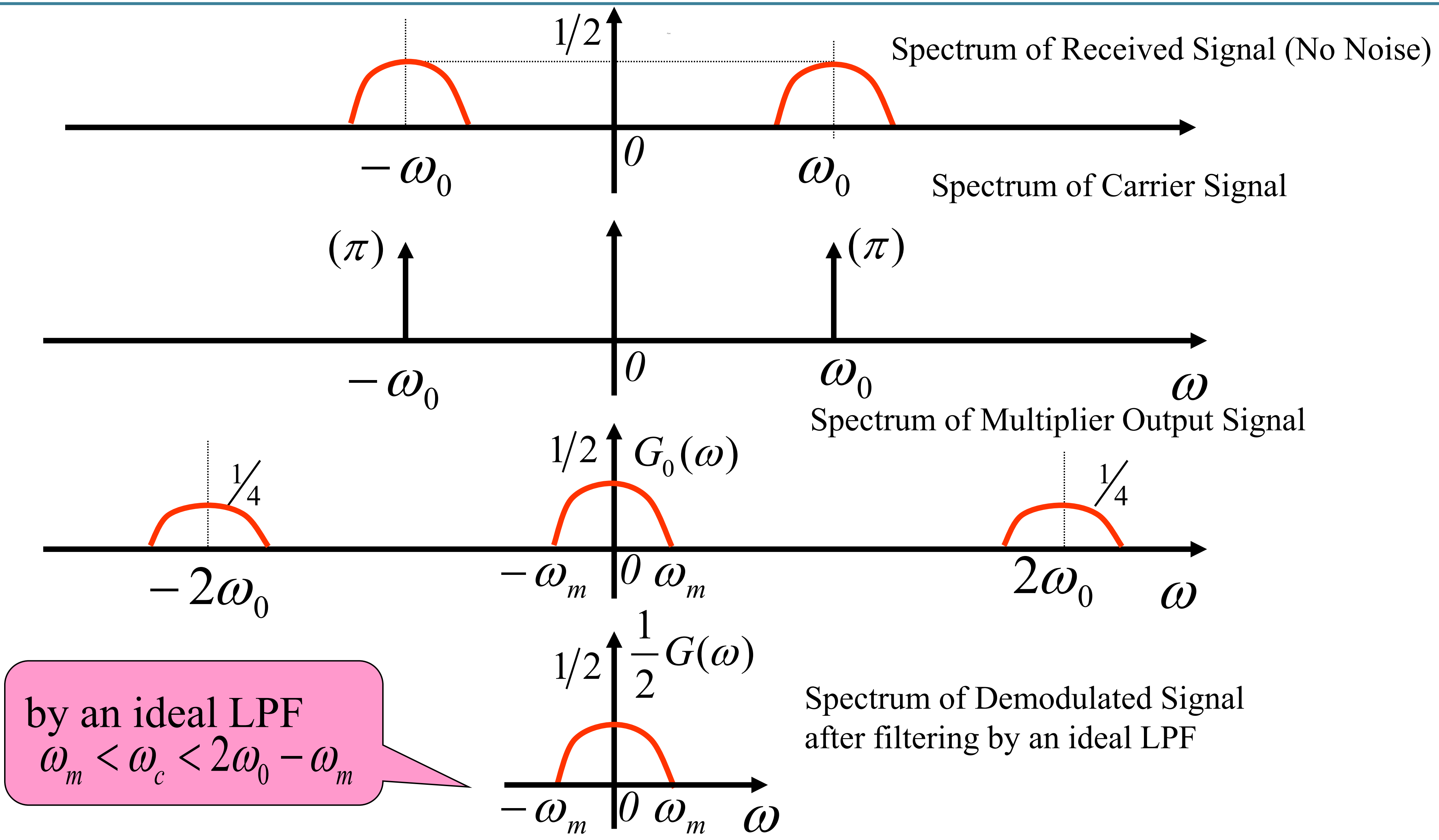


modulated signal $f(t) = g(t)\cos\omega_0 t$

$$F(\omega) = \frac{1}{2\pi} G(\omega) * [\pi\delta(\omega+\omega_0) + \pi\delta(\omega-\omega_0)]$$

$$= \frac{1}{2}[G(\omega+\omega_0) + G(\omega-\omega_0)]$$





Spectrum of Received Signal (No Noise)
1/2
$-\omega_0$
0
ω_0
Spectrum of Carrier Signal
(π)
(π)
$-\omega_0$
0
ω_0
ω
Spectrum of Multiplier Output Signal
1/2
$G_0(\omega)$
1/4
1/4
$-2\omega_0$
$-\omega_m$
0
ω_m
$2\omega_0$
ω
1/2
$\frac{1}{2}G(\omega)$
Spectrum of Demodulated Signal
after filtering by an ideal LPF
by an ideal LPF
$\omega_m < \omega_c < 2\omega_0 - \omega_m$
$-\omega_m$
0
ω_m
ω

Problem 5-1: Given the spectrum of a signal $f(t)$ shown in Fig.1, the signal $f_1(t)$ is obtained by multiplying $f(t)$ with $\cos(100t)$ and the signal $f_2(t)$ is obtained by multiplying $f_1(t)$ with $\cos(100t)$ again. Then the signal $f_2(t)$ passes through an ideal LPF with a cutoff frequency of ω_c to get the signal $r(t)$.

(1) Draw the spectrum $F_1(\omega)$ and $F_2(\omega)$ corresponding to the signals $f_1(t)$ and $f_2(t)$.

(2) If $r(t) = f(t)$, find the range of the cutoff frequency of the ideal LPF ω_c.

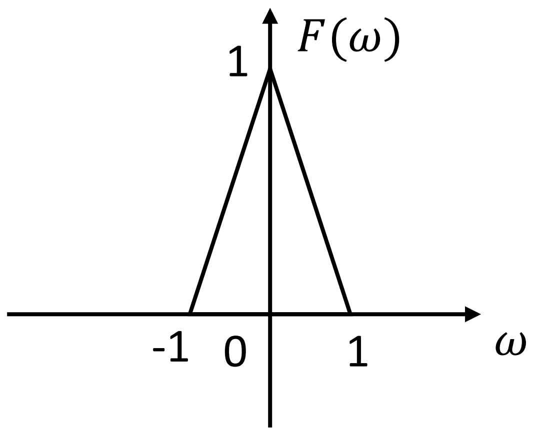


Fig. 1

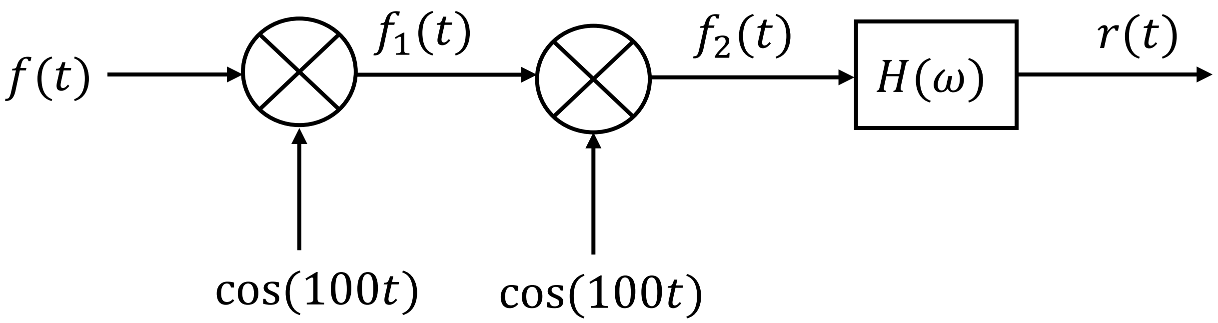
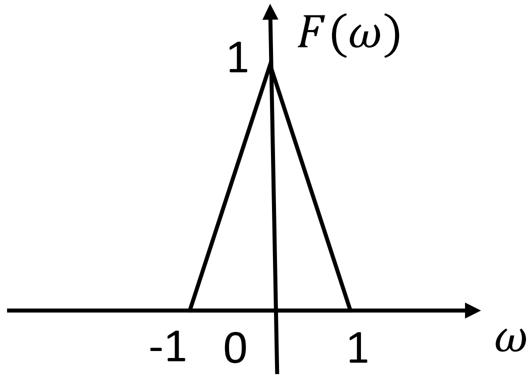
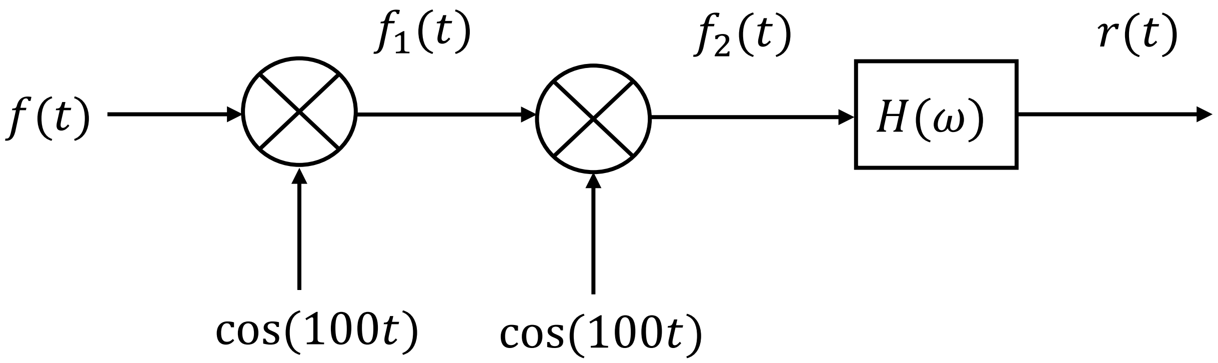


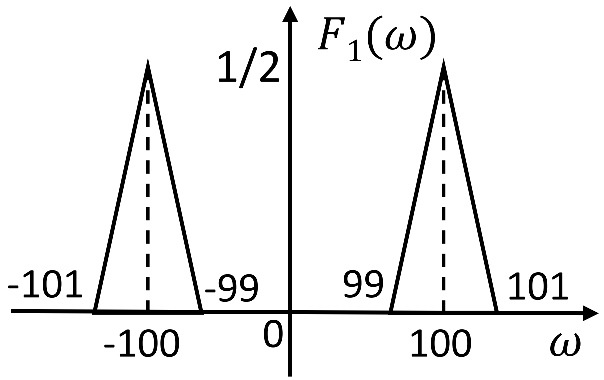
Fig. 2

Solution: (1) Draw the spectra of $F_1(\omega)$ and $F_2(\omega)$.

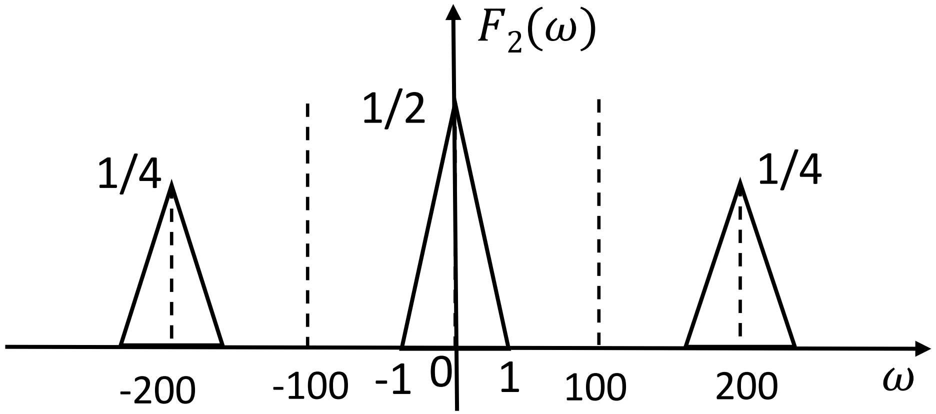




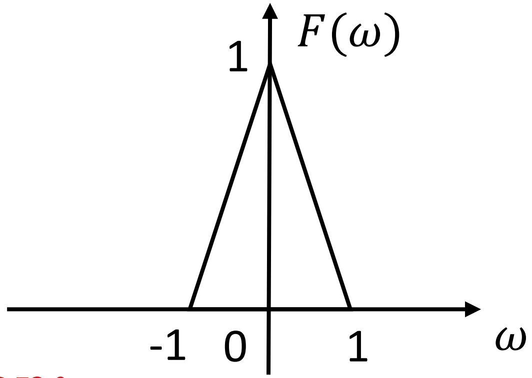
Spectrum of $F_1(\omega)$

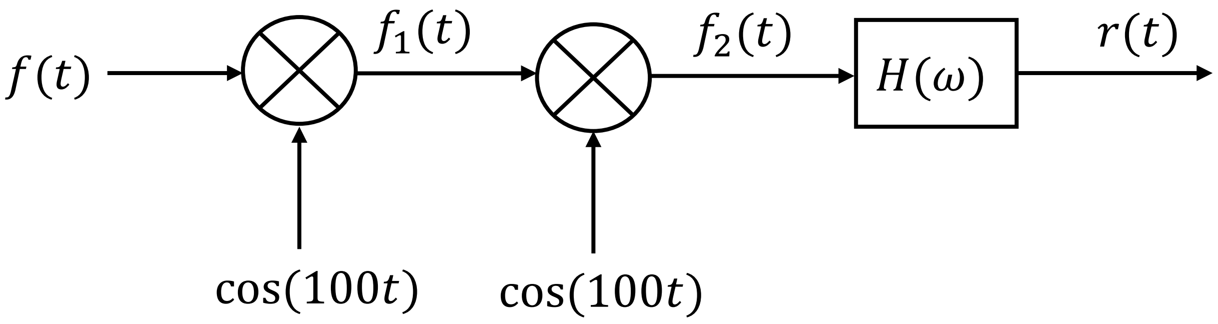


Spectrum of $F_2(\omega)$

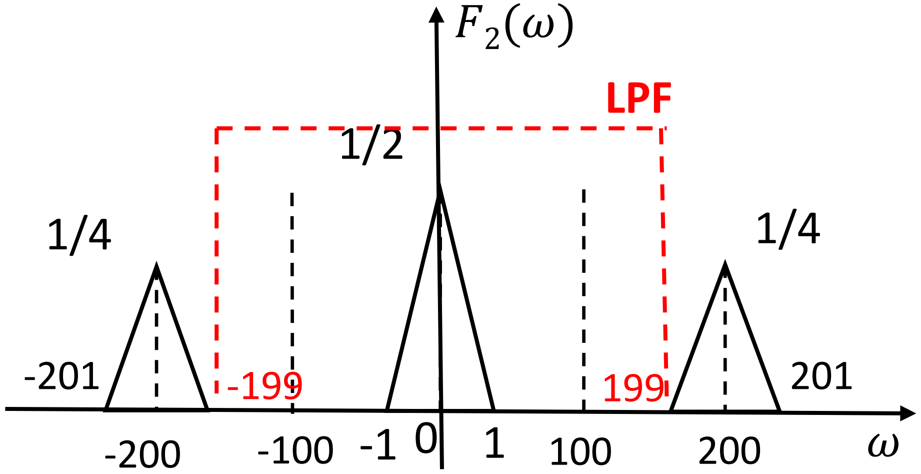


(2) If $r(t) = f(t)$, find the range of the cutoff frequency of the LPF ω_c.





Solution:



If $r(t) = f(t)$, we need retain the $\frac{1}{2}F(w)$ component in $F_2(w)$ (corresponding to $|w| < 1$; filter out the $\frac{1}{4}F(w \pm 200)$ components in $F_2(w)$ (corresponding to $|w| > 199$).

Therefore, the cutoff frequency w_c must satisfy:

$$1 < w_c < 199$$

单选题 1分

设置

Determine whether the following sequences are periodic or not.

$$(1)\ \cos\left(\frac{3}{7}n-\frac{\pi}{8}\right) \qquad (2)\ e^{j\frac{\pi}{8}n}$$

A Periodic; periodic

B Periodic; aperiodic

C Aperiodic; periodic

D Aperiodic; aperiodic

提交

Determination of Periodicity of Sinusoidal Sequence $x(n)=\sin(n\omega_0)$

① $\dfrac{2\pi}{\omega_0}=N$, N is a positive integer

$$\sin\left[\omega_0\left(n+N\right)\right]=\sin\left[\omega_0\left(n+\frac{2\pi}{\omega_0}\right)\right]=\sin\left(\omega_0 n+2\pi\right)=\sin\left(\omega_0 n\right)$$

The sinusoidal sequence is periodic, **with period *N***.

② $\dfrac{2\pi}{\omega_0}=\dfrac{N}{m},\dfrac{N}{m}$ is a rational number

$$\sin\left[\omega_0\left(n+N\right)\right]=\sin\left[\omega_0\left(n+m\frac{2\pi}{\omega_0}\right)\right]=\sin\left(\omega_0 n+m\cdot 2\pi\right)=\sin\left(\omega_0 n\right)$$

$\sin\left(\omega_0 n\right)$ is still periodic. **Period:** $m\dfrac{2\pi}{\omega_0}=N$

③ $\dfrac{2\pi}{\omega_0}$ is an irrational number

There is no value of N that satisfies $x\left(n+N\right)=x\left(n\right)$, so the sequence is **aperiodic**.

Problem 6-1: Determine whether the following sequence is a periodic signal. If yes, what is its period?

(1) $\cos\left(\frac{3}{7}n-\frac{\pi}{8}\right)$ (2) $e^{j\frac{\pi}{8}n}$

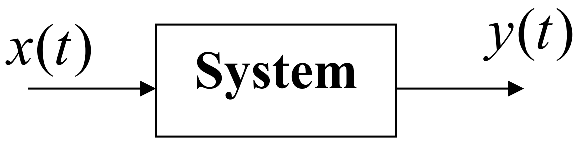
Solution:

(1) $\omega_0=\frac{3}{7}$, $\frac{2\pi}{\omega_0}=\frac{14\pi}{3}$ is an irrational number.

Thus, the sequence is **aperiodic**.

(2) $\omega_0=\frac{\pi}{8}$, $\frac{2\pi}{\omega_0}=16$ is a positive integer.

Thus, the sequence is **periodic**, with a period of **16**.



Linearity: Linearity includes superposition and homogeneity.

Time-invariant System: The parameters of the system do not change with time; if the excitation is delayed, the response is delayed by the same amount.

Causal System: A system where the output change does not precede the input change. That is, the response depends only on the current and previous excitations.

The **necessary and sufficient condition** for a discrete-time LTI system to be **a causal system** is:

$$h(n) = 0 \ (n < 0) \text{ or } \ h(n) = h(n)u(n)$$

The systems has input $x(n)$ and output $y(n)$. For $y(n) = 2x(n)u(n)$, determine whether the system is linear, time invariant and causal

A Linear, time-invariant, causal

B Linear, time-varying, causal

C Nonlinear, time-invariant, causal

D Nonlinear, time-varying, non-causal

提交

Problem 6-2: The systems that follow have input $x(n)$ and output $y(n)$. For each system, determine whether it is linear, time invariant and causal.

$$(1)\quad y(n)=2x(n)u(n) \qquad\qquad (2)\quad y(n)=\cos(x(n))$$

Solution: (1) **Linearity**:

Let: $x_1(n)\rightarrow y_1(n)=2x_1(n)u(n)$, $x_2(n)\rightarrow y_2(n)=2x_2(n)u(n)$

For any constants a,b

$$ax_1(n)+bx_2(n)\rightarrow 2[ax_1(n)+bx_2(n)]u(n)=ay_1(n)+by_2(n)$$

Thus, the system is **linear**.

Time Invariance:

$x(n)\rightarrow y(n)=2x(n)u(n)$

input $x(n-k)\rightarrow$ output $=2x(n-k)u(n)$

but $y(n-k)=2x(n-k)u(n-k)$

Thus, the system is **time-varying**.

Causality:

$y(n)$ only uses $x(n)$ at time n

Thus, the system is **causal**.

(2) $y(n)=\cos(x(n))$

Linearity:

$y_1(n)=\cos\left[x_1(n)\right]\quad y_2(n)=\cos\left[x_2(n)\right]$

$\cos\left[ax_1(n)+bx_2(n)\right]=\cos\left[ax_1(n)\right]\cos\left[bx_2(n)\right]-\sin\left[ax_1(n)\right]\sin\left[bx_2(n)\right]$

$ay_1(n)+by_2(n)=a\cos\left[x_1(n)\right]+b\cos\left[x_2(n)\right]\neq\cos\left[ax_1(n)+bx_2(n)\right]$

Thus, the system is **nonlinear**.

Time Invariance:

$x(n)\to y(n)=2x(n)u(n)$

input $x(n-k)\to$ output $=\cos(x(n-k))$

and $y(n-k)=\cos(x(n-k))$

Thus, the system is **time-invariant**.

Causality:

$y(n)$ only uses $x(n)$ at time n

Thus, the system is **causal**.

Complete Solution = Homogeneous Solution + Particular Solution

Total Response = Natural Response + Forced Response

Total Response = Zero-Input Response + Zero-State Response

$$y(n) = \underbrace{\sum_{k=1}^{N} C_k \alpha_k^n}_{\text{Natural Response}} + \underbrace{D(n)}_{\text{Forced Response}}$$

$$= \underbrace{\sum_{k=1}^{N} C_{zik} \alpha_k^n}_{\text{Zero-Input Response}} + \sum_{k=1}^{N} \underbrace{C_{zsk} \alpha_k^n + D(n)}_{\text{Zero-State Response}}$$

Zero-Input Response

➢Definition: The response generated solely by the initial state of the system (with the input signal being zero), denoted as $\boldsymbol{y_{zi}(n)}$.

➢Mathematical Model:

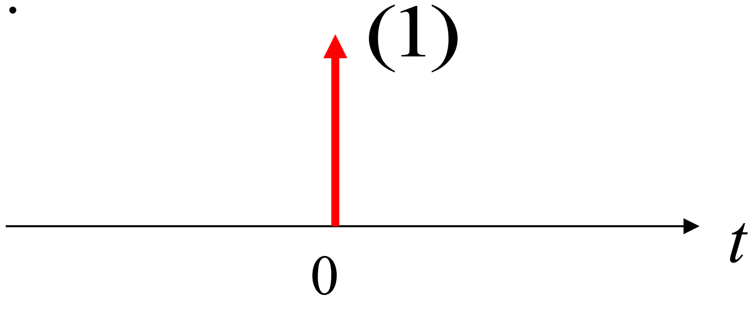
$$\sum_{k=0}^{N} a_k y(n-k) = 0$$

➢Solution Method: Determine the form of the zero-input response based on the characteristic roots of the difference equation, then solve for the **undetermined coefficients of the homogeneous solution** using **boundary conditions**.

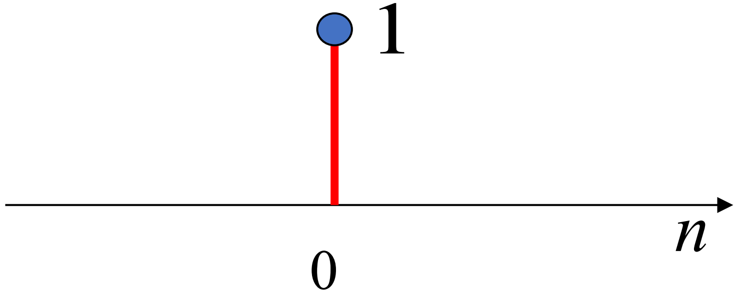
Unit Sample Response: It is the **zero-state response** of the system when the excitation is $\delta(n)$, denoted by $\boldsymbol{h(n)}$**.**

Note the difference between $\delta(n)$ and $\delta(t)$.

$$\begin{cases} \int_{-\infty}^{\infty} \delta(t)dt = 1 & (t=0) \\ \delta(t) = 0 & (t \neq 0) \end{cases}$$



$$\delta(n) = \begin{cases} 1 & n = 0 \\ 0 & n \neq 0 \end{cases}$$



Solution of Unit Sample Response: Convert the unit sample excitation signal into boundary conditions.

Convert $\delta(n)$ into boundary conditions, so finding the unit sample response is converted into finding **the homogeneous solution**, i.e., **the zero-input response when** $\boldsymbol{n>0}$**.**

单选题 1分

设置

Consider an LTI system described by the following difference equation:

$$y(n)-\frac{5}{6}y(n-1)+\frac{1}{6}y(n-2)=x(n)$$

The initial states of the system are $y(-1)=0$， $y(-2)=3$. The zero-input response of the system is ( ).

A $-1.5\times 2^{-n}+3^{-n} \quad (n\geq 0)$

B $0.3\times 2^{-n}+0.2\times 3^{-n} \quad (n\geq 0)$

C $0.2\times 2^{-n}+0.3\times (-3)^{-n} \quad (n\geq 0)$

D $1.5\times 2^{-n}+3^{-n} \quad (n\geq 0)$

提交

Problem 6-3: Consider an LTI system described by the following difference equation:

$$y(n)-\frac{5}{6}y(n-1)+\frac{1}{6}y(n-2)=x(n)$$

The initial states of the system are $y(-1)=0$， $y(-2)=3$.

(1) Find the zero-input response of the system.

(2) Find the unit sample response $h(n)$.

Solution: (1) The characteristic equation of the system is: $\alpha^2-\frac{5}{6}\alpha+\frac{1}{6}=0$

The characteristic roots are $\alpha_1=\frac{1}{2},\alpha_2=\frac{1}{3}$

Assume the zero-input response of the system is: $y_{zi}(n)=C_1 2^{-n}+C_2 3^{-n}$

$$\begin{aligned} y(-1)&=2C_1+3C_2=0 \\ y(-2)&=4C_1+9C_2=3 \end{aligned} \quad \rightarrow C_1=-1.5, \quad C_2=1$$

$$\therefore y_{zi}(n)=-1.5\times 2^{-n}+3^{-n} \quad (n\geq 0)$$

(2) The difference equation is $y(n)-\frac{5}{6}y(n-1)+\frac{1}{6}y(n-2)=x(n)$

Thus, we have $h(n)-\frac{5}{6}h(n-1)+\frac{1}{6}h(n-2)=\delta(n)$

When $n<0$, $h(n)=0$

When $n=0$, $h(0)=\frac{5}{6}h(-1)-\frac{1}{6}h(-2)+\delta(0)=1$

When $n>0$, $h(n)-\frac{5}{6}h(n-1)+\frac{1}{6}h(n-2)=0$

Homogeneous solution: $h(n)=C_1 2^{-n}+C_2 3^{-n}$

$$h(0)=C_1+C_2=1$$

$$h(-1)=2C_1+3C_2=0$$

$\rightarrow C_1=3$, $C_2=-2$

The unit sample response is: $h(n)=(3\times 2^{-n}-2\times 3^{-n})u(n)$

Operation Method of Convolution Sum

$$y(n) = x(n) * h(n) = \sum_{m=-\infty}^{\infty} x(m)h(n-m)$$

The graphical method of convolution sum consists of five steps:

(1) Variable Substitution: Change the independent variable in $\boldsymbol{x(n)}$ and $\boldsymbol{h(n)}$ from $\boldsymbol{n}$ to $\boldsymbol{m}$, where $\boldsymbol{m}$ becomes the independent variable of the function.

(2) Reversing: Fold one of the signals (usually the simpler one)

$$h(m) \xrightarrow{\text{Reverse}} h(-m)$$

(3) Shifting: Shift the folded signal

$$h(-m) \xrightarrow{\text{right shift by } n} h(-(m-n)) = h(n-m)$$

(4) Multiplying: Multiply $\boldsymbol{x(m)}$ by $\boldsymbol{h(n\text{-}m)}$

(5) Summing: Sum the product graph.

Given a discrete-time LTI system, its unit sample response ***h*(*n*)** has non-zero values only in the interval $N_0 \le n \le N_1$, and the input ***x*(*n*)** has non-zero values only in the interval of $N_2 \le n \le N_3$. Then, the zero-state response has non-zero values only in the interval of $N_4 \le n \le N_5$, where

$$\boxed{N_4 = N_0 + N_2 \quad N_5 = N_1 + N_3}$$

Assume the unit sample response ***h*(*n*)** has non-zero values only in $N_0 \le n \le N_1$, with sequence length $L_1 = N_1 - N_0 + 1$. The input ***x*(*n*)** has non-zero values only in $N_2 \le n \le N_3$, with sequence length $L_2 = N_3 - N_2 + 1$.

The zero-state response ***y*(*n*)** has non-zero values only in $N_0 + N_2 \le n \le N_1 + N_3$, with **sequence length *L*$_3$**:

$$\boxed{L_3 = L_1 + L_2 - 1}$$

For continuous-time signals, the length of the convolution integral is $L_3 = L_1 + L_2$.

单选题 1分

设置

Calculate the convolution sum of $x(n)=\{1,\ 2,\ \overset{\downarrow}{0},\ 3,\ 2\}$ and $h(n)=\{1,\ \overset{\downarrow}{4},\ 2,\ 3\}$

A $y(n)=\{1,6,\overset{\downarrow}{10},10,20,14,13,6\}$

B $y(n)=\{1,8,\overset{\downarrow}{10},10,21,14,13,6\}$

C $y(n)=\{1,6,10,\ \overset{\downarrow}{10},20,14,13,6\}$

D $y(n)=\{1,8,10,\overset{\downarrow}{10},21,14,13,6\}$

提交

Problem 6-4: Given $x(n)=\{1, 2, \overset{\downarrow}{0}, 3, 2\}$, $h(n)=\{1, \overset{\downarrow}{4}, 2, 3\}$.

Find the convolution $y(n)=x(n)*h(n)$

Solution:

$$
\begin{array}{ccccccccc}
 & & & x(n): & 1 & 2 & 0 & 3 & 2 \\
 & & & h(n): & & 1 & 4 & 2 & 3 \\
\hline
 & & & & 3 & 6 & 0 & 9 & 6 \\
 & & & 2 & 4 & 0 & 6 & 4 & \\
 & & 4 & 8 & 0 & 12 & 8 & & \\
 & 1 & 2 & 0 & 3 & 2 & & & \\
\hline
1 & 6 & 10 & \underset{\uparrow}{10} & 20 & 14 & 13 & 6 &
\end{array}
$$

$y(n)=\{1, 6, 10, \overset{\downarrow}{10}, 20, 14, 13, 6\}$

Finding Zero-State Response Using Convolution Sum

Decompose the excitation signal into a linear combination of unit sample sequences. Using the properties of linear time-invariant (LTI) systems, find the response of each sample sequence acting on the system individually, and superimpose these responses to obtain the **zero-state response** under any excitation.

Any excitation signal can be expressed as a weighted sum of unit samples：

$$x(n) = \sum_{m=-\infty}^{\infty} x(m)\delta(n-m)$$

$\delta(n) \Rightarrow h(n)$ From time-invariance: $\delta(n-m) \Rightarrow h(n-m)$

From linearity： $y(n) = T\left\{\sum_{m=-\infty}^{\infty} x(m)\delta(n-m)\right\} = \sum_{m=-\infty}^{\infty} x(m)h(n-m)$

$$= \sum_{m=-\infty}^{\infty} h(m)x(n-m) = x(n) * h(n)$$

The zero-state response of a discrete-time LTI system is equal to the convolution sum of the excitation signal and the unit sample response.

Problem 6-5: Consider the LTI system in **Problem 6-3** with the following difference equation:

$$y(n)-\frac{5}{6}y(n-1)+\frac{1}{6}y(n-2)=x(n)$$

If the excitation signal is $\boldsymbol{x(n)=u(n)}$ (unit step sequence), find the zero-state response of the system using the convolution sum.

Solution: The unit sample response has been found as: $h(n)=(3\times 2^{-n}-2\times 3^{-n})u(n)$

$$y_{sz}(n)=h(n)*x(n)=\sum_{m=-\infty}^{\infty}h(m)x(n-m)$$

$$=\sum_{m=-\infty}^{\infty}(3\times 2^{-m}-2\times 3^{-m})u(m)u(n-m)$$

$$=\sum_{m=0}^{n}(3\times 2^{-m}-2\times 3^{-m})$$

$$=\sum_{m=0}^{n}\left[6\times(1-2^{-n-1})-3\times(1-3^{-n-1})\right]$$

$$=3-3\times 2^{-n}+3^{-n}\quad (n\geq 0)$$

For a continuous causal signal $\boldsymbol{x(t)}$ sampled at interval $\boldsymbol{T}$, the sampled signal $\boldsymbol{x_s(t)}$ is

$$x_s(t) = x(t)\delta_T(t) = \sum_{n=0}^{\infty} x(nT)\delta(t-nT)$$

Unilateral Laplace Transform

$$X_s(s) = \int_0^{\infty} x_s(t)e^{-st}dt = \sum_{n=0}^{\infty} x(nT)\int_0^{\infty}\delta(t-nT)e^{-st}dt = \sum_{n=0}^{\infty} x(nT)e^{-snT}$$

Let $z = e^{sT}$,where **z is a complex variable**. The relationship between the Laplace transform of the sampled signal and the z-transform of the sequence is as follows:

$$\boxed{X(z)\Big|_{z=e^{sT}} = X_s(s)}$$

Usually let T=1, $z = e^s$

Unilateral z-Transform

$$X(z) = \sum_{n=0}^{\infty} x(n)z^{-n}$$

Bilateral z-Transform

$$X(z) = \sum_{n=-\infty}^{\infty} x(n)z^{-n}$$

The z-transform of a sequence is **a power series of the complex variable** z^{-1} , and **its coefficients are** $\boldsymbol{x(n)}$**.**

单选题 1分

设置

Given a z-transform $X(z) = \frac{-3z^{-1}}{2-5z^{-1}+2z^{-2}}$ $(\frac{1}{2} < |z| < 2)$, the corresponding sequence $x(n)$ is ( ).

A Right-sided

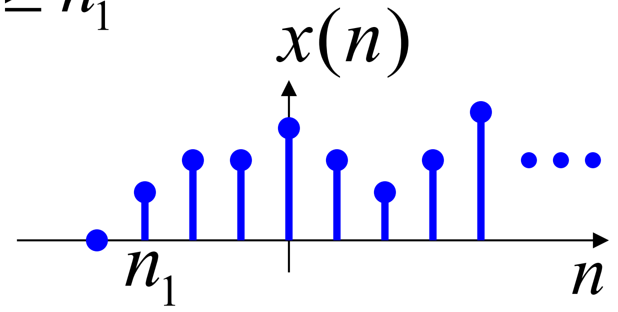
B Left-sided

C Double-sided

D Of finite length

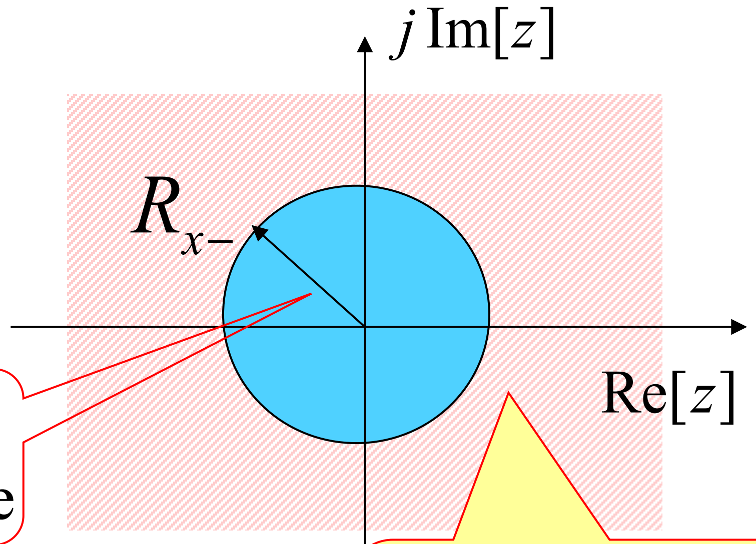
提交

1. Right-Sided Sequence: A sequence that has non-zero finite values only in the interval $n \geq n_1$



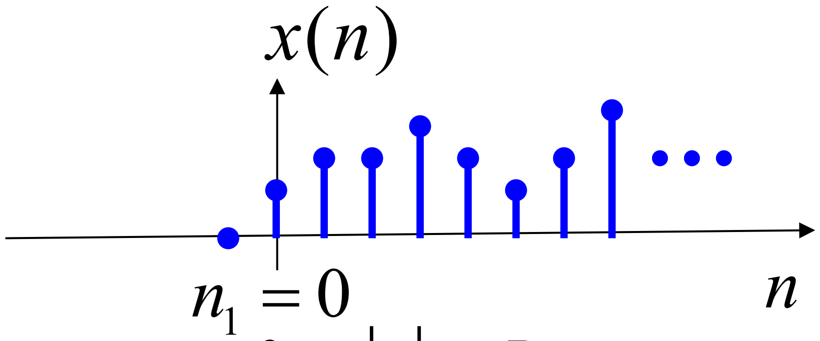
$$X(z) = \sum_{n=n_1}^{\infty} x(n) z^{-n} \qquad n_1 \leq n \leq \infty$$

$n_1 < 0, \quad R_{x-} < |z| < \infty$ (The ROC does not include $|z| = \infty$, because ∞^{-n_1} is infinite)



Radius of Convergence

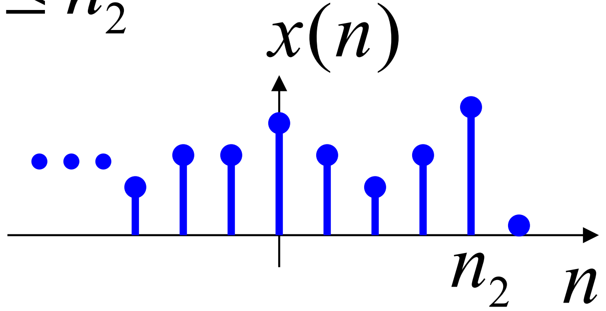
The ROC is outside the circle. If $\boldsymbol{n_1<0}$, $|z| = \infty$ is not included.



$n_1 = 0, \quad |z| > R_{x-}$

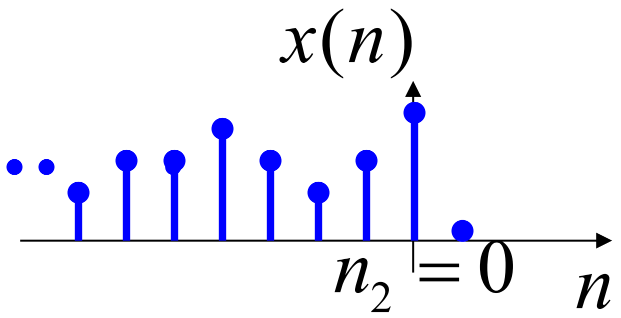
A **causal sequence** is a special case of a right-sided sequence, and its ROC is $|z| > R_{x-}$

2. Left-Sided Sequence: A sequence $x(n)$ that has non-zero finite values only in the interval $n \le n_2$

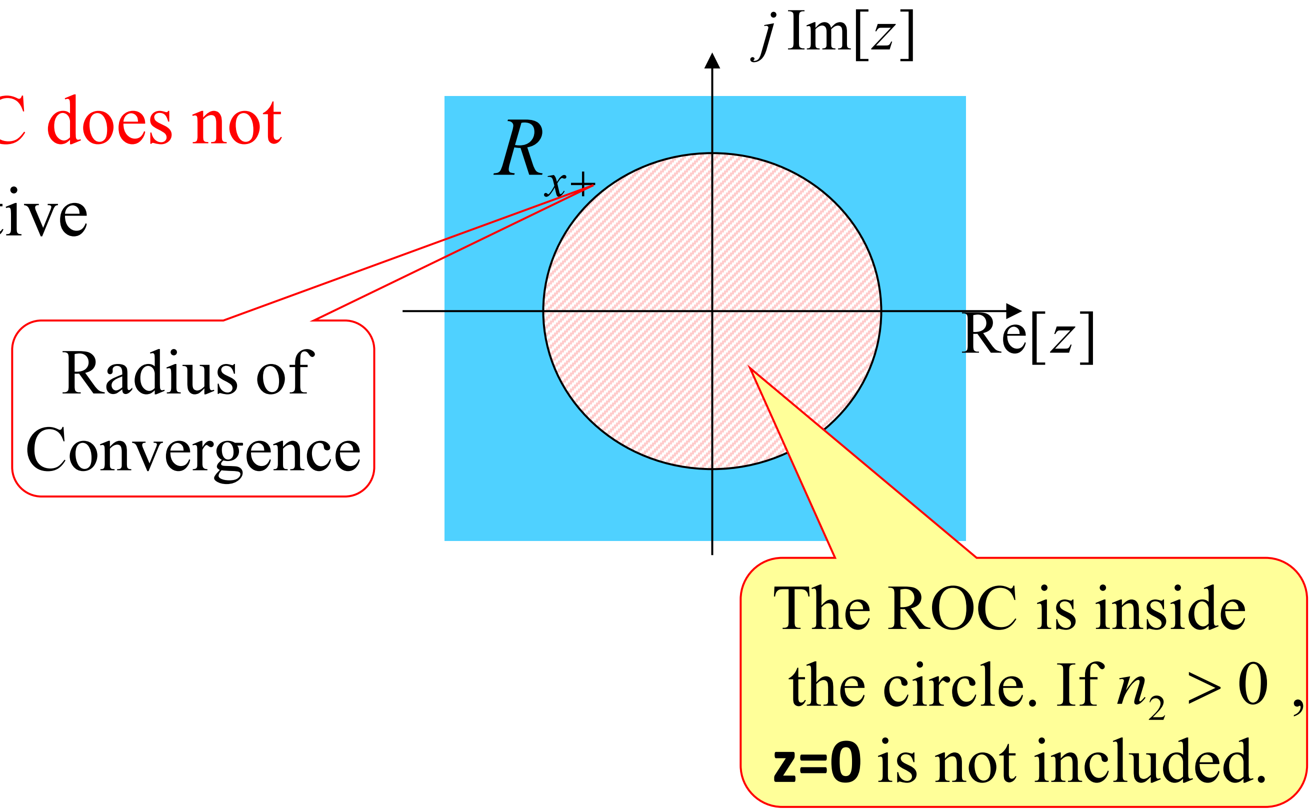


$$X(z) = \sum_{n=-\infty}^{n_2} x(n) z^{-n} \qquad -\infty \le n \le n_2$$

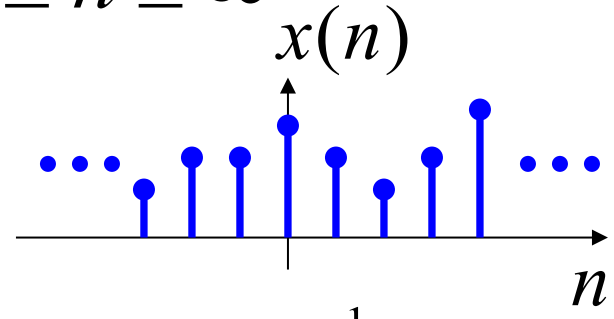
$n_2 > 0, \quad 0 < |z| < R_{x+}$ （The ROC does not include $z = 0$, because the negative power of 0 is meaningless）



$n_2 = 0, \quad |z| < R_{x+}$



3. Double-Sided Sequence: A sequence that has non-zero finite values in the interval $-\infty \leq n \leq \infty$

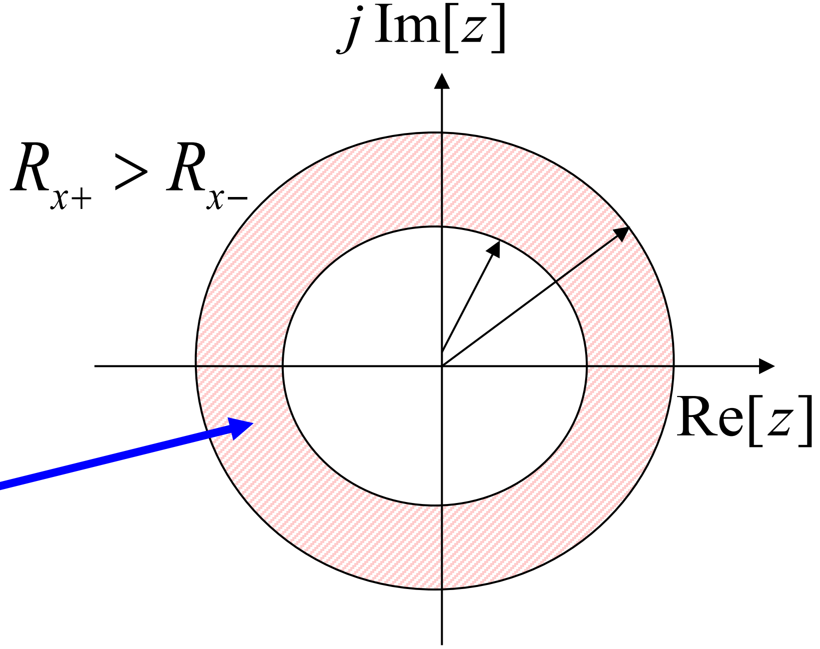


$$X(z)=\sum_{n=-\infty}^{\infty} x(n) z^{-n} \qquad -\infty \leq n \leq \infty$$

$$X(z)=\sum_{n=-\infty}^{-1} x(n) z^{-n}+\sum_{n=0}^{\infty} x(n) z^{-n}$$

Converges inside the circle $|z|<R_{x+}$

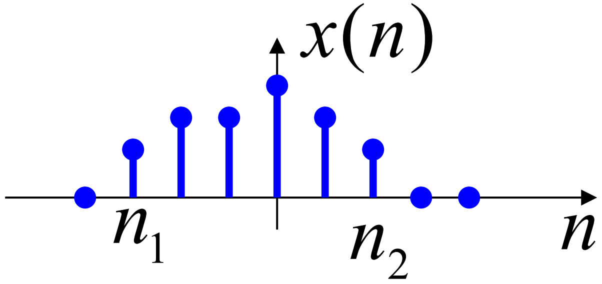
Converges outside the circle $|z|>R_{x-}$



$R_{x+} > R_{x-}$ There is an annular ROC

$R_{x+} < R_{x-}$ No ROC

4. Finite-Length Sequence: A sequence that has non-zero finite values in the finite interval $n_1 \le n \le n_2$



$$X(z) = \sum_{n=n_1}^{n_2} x(n) z^{-n} \qquad n_1 \le n \le n_2$$

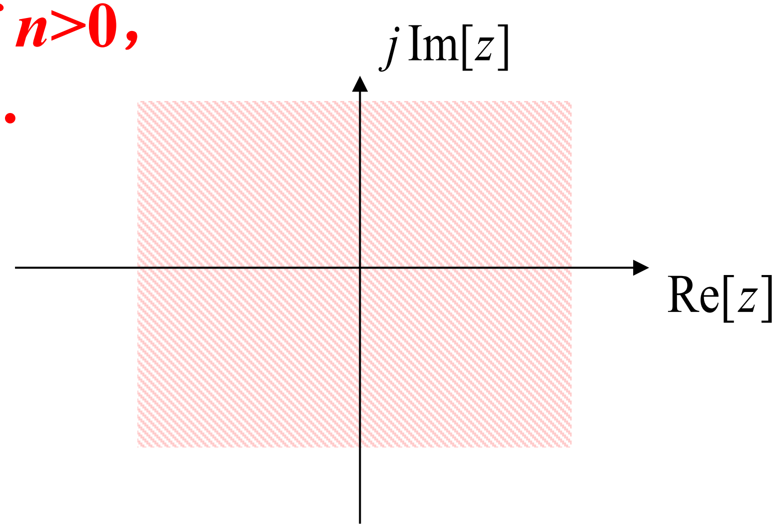
Divided into three cases:

$n_1 < 0, n_2 > 0 \quad 0 < |z| < \infty$

$n_1 \ge 0, n_2 > 0 \quad 0 < |z| \le \infty$

$n_1 < 0, n_2 \le 0 \quad 0 \le |z| < \infty$

0^{-n} is meaningless if n>0，
∞^{-n} is infinite if n<0.



The ROC is at least the entire z-plane but 0 and ∞ (If pole-zero cancellation occurs, the ROC expands to the entire z-plane).

$$X(z)=ZT[x(n)]=\sum_{n=-\infty}^{\infty}x(n)z^{-n}$$

Partial Fraction Expansion Method: $X(z)=\dfrac{b_0+b_1z+\cdots b_{r-1}z^{r-1}+b_rz^r}{a_0+a_1z+\cdots+a_{k-1}z^{k-1}+a_kz^k}$

***X*(z) has only first-order poles**

$$X(z)=A_0+\sum_{m=1}^{M}\frac{A_mz}{z-z_m}=A_0+\frac{A_1z}{z-z_1}+\cdots+\frac{A_Mz}{z-z_M} \qquad A_0=X(z)\big|_{z=0}=\frac{b_0}{a_0}$$

Let z_m be a first-order pole of $X(z)$ $\quad A_m=\left[(z-z_m)\dfrac{X(z)}{z}\right]_{z=z_m} \qquad X(z)=A_0+\displaystyle\sum_{m=1}^{M}\frac{zA_m}{z-z_m}$

or $\quad X(z)=A_0+\displaystyle\sum_{m=1}^{M}\frac{A_m}{1-z_mz^{-1}}=A_0+\frac{A_1}{1-z_1z^{-1}}+\cdots+\frac{A_M}{1-z_Mz^{-1}} \qquad A_m=\left[\left(1-z_mz^{-1}\right)X(z)\right]_{z=z_m}$

Inverse Transform: $x(n)=A_0\delta(n)+\displaystyle\sum_{m=1}^{M}A_mz_m^n=A_0\delta(n)+A_1z_1^n+\cdots+A_Mz_M^n$

Problem 7-1: Find the inverse Z-transform of $X(z)=\frac{-3z^{-1}}{2-5z^{-1}+2z^{-2}}$ $(\frac{1}{2}<|z|<2)$.

Solution: **Step 1**: Partial fractional decomposition

$$X(z)=\frac{-3z^{-1}}{2-5z^{-1}+2z^{-2}}=\frac{A_1}{1-\frac{1}{2}z^{-1}}+\frac{A_2}{1-2z^{-1}}$$

$$A_1=(1-\frac{1}{2}z^{-1})X(z)\Big|_{z=\frac{1}{2}}=1 \qquad A_2=(1-2z^{-1})X(z)\Big|_{z=2}=-1$$

$$X(z)=\frac{1}{1-\frac{1}{2}z^{-1}}-\frac{1}{1-2z^{-1}}$$

Step 2: Determine the inverse Z-transform based on ROC

For $\dfrac{1}{1-\frac{1}{2}z^{-1}}$, ROC $|z| > \frac{1}{2}$ corresponds to a right-sided sequence: $\left(\dfrac{1}{2}\right)^n u(n)$

For $\dfrac{1}{1-2z^{-1}}$, ROC $|z| < 2$ corresponds to a left-sided sequence: $-2^n u(-n-1)$

Combine the results: $$x(n) = \left(\frac{1}{2}\right)^n u(n) + 2^n u(-n-1)$$

This is a double-sided sequence.

单选题 1分

设置

Given that the Z-transform of sequence $x(n)$ is $X(z)=\frac{z}{z-\frac{1}{2}}$ $(|z|>\frac{1}{2})$， the Z-transform of the sequence $y(n)=3^n x(n)$ and its ROC is ( ).

A $Y(z)=\dfrac{z}{z-\frac{1}{6}}\left(|z|>\frac{3}{2}\right)$

C $Y(z)=\dfrac{z}{z-\frac{3}{2}}\left(|z|>\frac{1}{6}\right)$

B $Y(z)=\dfrac{z}{z-\frac{3}{2}}\left(|z|>\frac{3}{2}\right)$

D $Y(z)=\dfrac{z}{z-\frac{1}{6}}\left(|z|>\frac{1}{6}\right)$

提交

z-Domain Scaling Property

$$ZT[x(n)] = X(z) \quad (R_{x1} < |z| < R_{x2})$$

$$ZT[a^n x(n)] = X\left(\frac{z}{a}\right) \quad (R_{x1} < \left|\frac{z}{a}\right| < R_{x2} \Rightarrow |a| R_{x1} < |z| < |a| R_{x2}) \qquad |a| > 1$$

$$ZT[a^{-n} x(n)] = X(az) \quad (R_{x1} < |az| < R_{x2} \Rightarrow \frac{R_{x1}}{|a|} < |z| < \frac{R_{x2}}{|a|}) \qquad |a| > 1$$

$$ZT[(-1)^n x(n)] = X(-z) \quad (R_{x1} < |z| < R_{x2})$$

$$ZT[(-1)^n u(n)] = z/(z+1) \quad (|z| > 1)$$

Problem 7-2: Given that the Z-transform of sequence $x(n)$ is $X(z) = \frac{z}{z-\frac{1}{2}}$ $(|z| > \frac{1}{2})$, find the Z-transform of the sequence $y(n) = 3^n x(n)$ and its ROC.

Solution: By the z-domain scaling property, $\mathcal{Z}[a^n u(n)] = X\left(\frac{z}{a}\right), |z| > |a| r$

$$a = 3 \qquad Y(z) = X\left(\frac{z}{3}\right) = \frac{z/3}{z/3 - 1/2} = \frac{z}{z - \frac{3}{2}} \qquad |z| > \frac{3}{2}$$

System Function

$$\sum_{k=0}^{N} a_k y(n-k) = \sum_{r=0}^{M} b_r x(n-r)$$

If the initial state of the system is **$y(l)=0$ ($-N \le l \le -1$)**， the system is in the zero state, and the excitation **$x(n)$** is a causal sequence, then:

$$\sum_{k=0}^{N} a_k z^{-k} Y(z) = \sum_{r=0}^{M} b_r z^{-r} X(z)$$

System Function:

$$\boxed{H(z) = \frac{\sum_{r=0}^{M} b_r z^{-r}}{\sum_{k=0}^{N} a_k z^{-k}}}$$

$$Y_{zs}(z) = X(z)H(z)$$

$$y_{zs}(n) = ZT^{-1}[X(z)H(z)]$$

$$H(z) = \sum_{n=0}^{\infty} h(n) z^{-n}$$

Z-Domain Condition for Stability

$$H(z)=\sum_{n=-\infty}^{\infty} h(n)z^{-n},\ \text{its existence requires}: \sum_{n=-\infty}^{\infty}\left|h(n)z^{-n}\right|<\infty$$

when $|z|=1, \sum_{n=-\infty}^{\infty}|h(n)|<\infty$, just satisfies the condition for system stability.

Therefore, for a **stable system**, the ROC of $H(z)$ should **include the unit circle** (regardless of whether the system is causal).

For a **causal stable system**, the ROC is $|z|>a\ (a<1)$, which means **all poles lie inside the unit circle**.

For the LTI system with the following difference equation:

$$y(n) - \frac{5}{6}y(n-1) + \frac{1}{6}y(n-2) = x(n)$$

Determine if the system is stable or not.

A Stable

B Unstable

提交

Problem 7-3: For the LTI system in **Problems 6-3 and 6-5** with the following difference equation:

$$y(n)-\frac{5}{6}y(n-1)+\frac{1}{6}y(n-2)=x(n)$$

(1) Find the system function $H(z)$ and determine the stability of the system.

(2) Draw the system block diagram.

(3) If the excitation is $x(n)=u(n)$, find the zero-state response $y_{zs}(n)$ using the convolution theorem.

Solution: (1) Find the system function.

Take the z transform of both sides of the equation.

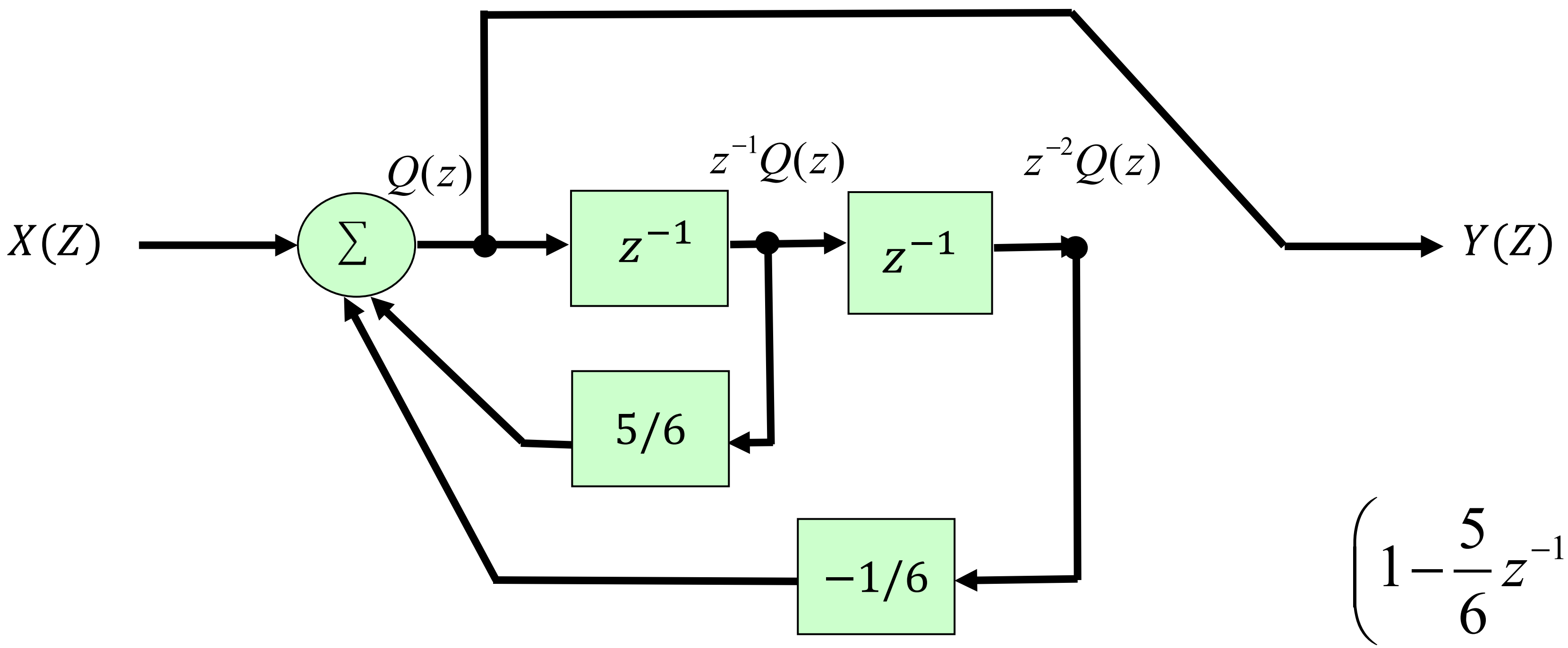
$$Y(z)-\frac{5}{6}z^{-1}Y(z)+\frac{1}{6}z^{-2}Y(z)=X(z)$$

$$H(z)=\frac{Y(z)}{X(z)}=\frac{1}{1-\frac{5}{6}z^{-1}+\frac{1}{6}z^{-2}}=\frac{1}{(1-\frac{1}{3}z^{-1})(1-\frac{1}{2}z^{-1})}\quad\left(z>\frac{1}{2}\right)\qquad z_1=\frac{1}{2}\qquad z_2=\frac{1}{3}$$

Both poles of $H(z)$ lie inside the unit circle. Thus, the system is **stable**.

(2) Draw the system block diagram.

$$H(z)=\frac{1}{1-\frac{5}{6}z^{-1}+\frac{1}{6}z^{-2}}=\frac{Y(z)}{X(z)}$$



$$\left(1-\frac{5}{6}z^{-1}+\frac{1}{6}z^{-2}\right)Q(z)=X(z)$$

$$Q(z)=Y(z)$$

(3) Find the zero-state response

$$Y_{zs}(z)=X(z)H(z)=\frac{z}{z-1}\cdot\frac{1}{(1-\frac{1}{2}z^{-1})(1-\frac{1}{3}z^{-1})}=\frac{1}{(1-z^{-1})(1-\frac{1}{2}z^{-1})(1-\frac{1}{3}z^{-1})}$$

$$Y_{zs}(z)=\frac{A_1}{1-z^{-1}}+\frac{A_2}{1-\frac{1}{2}z^{-1}}+\frac{A_3}{1-\frac{1}{3}z^{-1}}$$

$$A_1=(1-z^{-1})Y(z)\Big|_{z=1}=3 \qquad A_2=(1-\frac{1}{2}z^{-1})Y(z)\Big|_{z=\frac{1}{2}}=-3$$

$$A_3=(1-\frac{1}{3}z^{-1})Y(z)\Big|_{z=\frac{1}{3}}=1$$

$$Y_{zs}(z)=3\cdot\frac{1}{1-z^{-1}}-3\cdot\frac{1}{1-\frac{1}{2}z^{-1}}+\frac{1}{1-\frac{1}{3}z^{-1}}\quad(|z|>1) \qquad y_{zs}(n)=\left(3-3\times2^{-n}+3^{-n}\right)u(n)$$

Same as the solution to **Problem 6-5**.

Frequency Response of Discrete-Time Systems

The frequency response of a discrete-time system is the **discrete-time Fourier transform (DTFT) of the system's unit sample response**:

$$H(e^{j\omega}) = \sum_{n=-\infty}^{\infty} h(n) e^{-j\omega n}$$

Physical significance: $H(e^{j\omega})$ describes how each harmonic component is modified, with ***h*(*n*)** serving as the weighting coefficient.

Since $h(n)$ is generally a real sequence, $|H(e^{j\omega})|$ is an even function, and $\varphi(\omega)$ is an odd function.

$H\left(\mathrm{e}^{j\omega}\right)$ is a periodic function with period 2π.

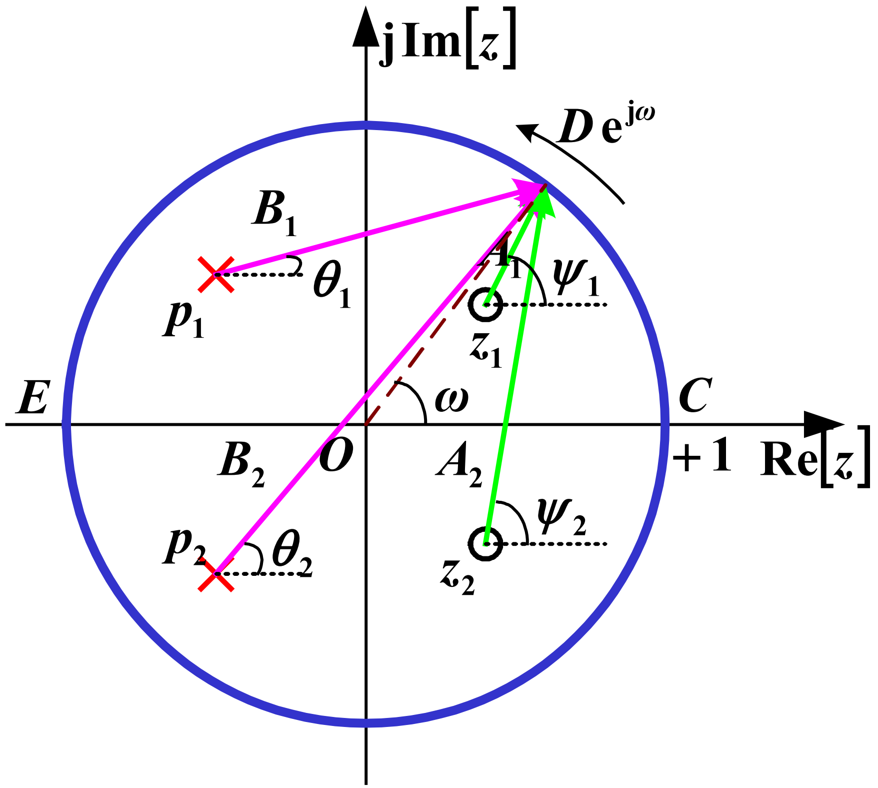
Geometric Determination of the Frequency Response Characteristic

$$H(z)=\frac{\prod_{r=1}^{M}(z-z_r)}{\prod_{k=1}^{N}(z-p_k)}$$

$$H\left(e^{j\omega}\right)=\frac{\prod_{r=1}^{M}\left(e^{j\omega}-z_r\right)}{\prod_{k=1}^{N}\left(e^{j\omega}-p_k\right)}=\left|H\left(e^{j\omega}\right)\right|e^{j\varphi(\omega)}$$

令 $e^{j\omega}-z_r=A_r\,e^{j\psi_r}$

$e^{j\omega}-p_k=B_k\,e^{j\theta_k}$



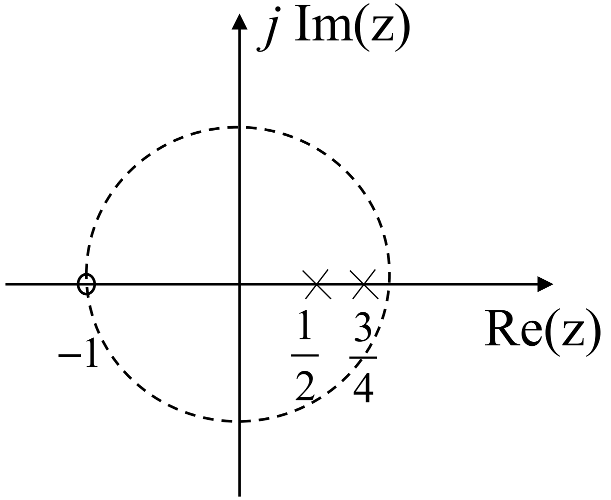
$$\text{Magnitude}-\text{frequency response}\left|H\left(e^{j\omega}\right)\right|=\frac{\prod_{r=1}^{M}A_r}{\prod_{k=1}^{N}B_k}\qquad \text{Phase}-\text{frequency response}\,\varphi(\omega)=\sum_{r=1}^{M}\psi_r-\sum_{k=1}^{N}\theta_k$$

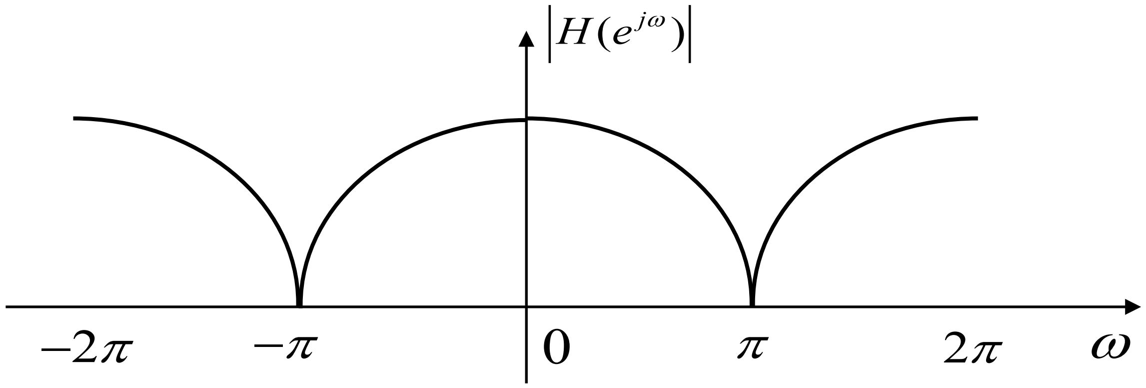
Problem 7-4: A discrete-time causal LTI system has the following system function:

$$H(z) = \frac{z+1}{z^2 - \frac{5}{4}z + \frac{3}{8}}$$

Draw the zero-pole plot of the system function, roughly sketch the magnitude-frequency response of the system based on this plot, and determine the filter type of the system.

Solution:





This is a low-pass filter.

- To analyze complex systems (e.g., **nonlinear, time-varying, multi-input multi-output** systems), the **state variable description method** (or **internal method**) is used. It describes the system's internal variables via state variables, and connects input/output variables through these state variables to characterize the system's external behavior.
- **State variable**: The **minimal** set of variables that describe the system's state. It essentially reflects changes in the system's internal energy storage state.
- **state：** If the system's state variables at $t = t_0$ and the input for $t \geq t_0$ are known, the system's state and output at any $t \geq t_0$ can be fully determined. **An *n*th-order system has *n* initial states.**

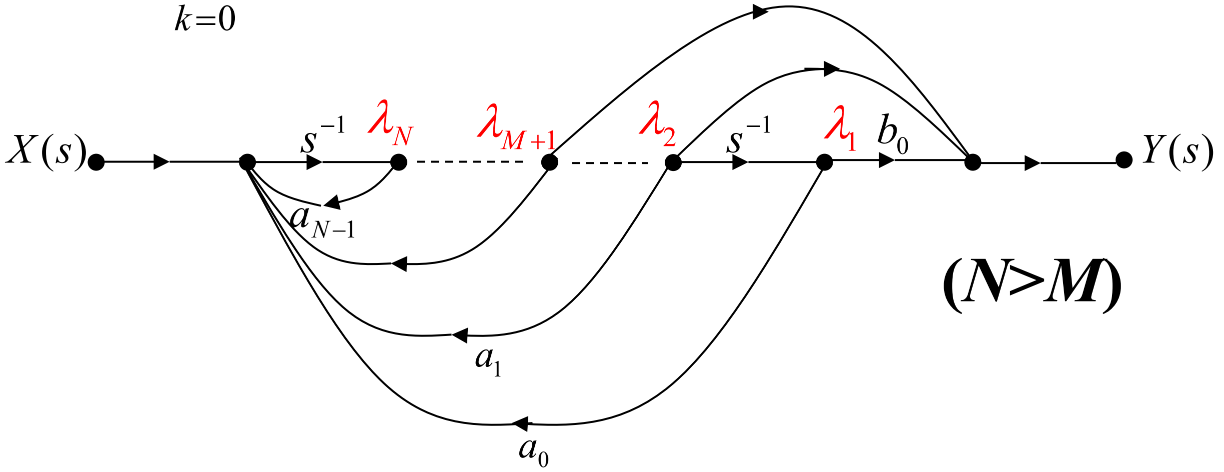
Formulation of State Equations and Output Equations for Continuous-Time Systems

The differential equation and system function of the system are as follows:

$$\frac{d^N y(t)}{dt^N} - \sum_{k=0}^{N-1} a_k \frac{d^k y(t)}{dt^k} = \sum_{r=0}^{M} b_r \frac{d^r x(t)}{dt^r}$$

$$H(s) = \frac{\sum_{r=0}^{M} b_r s^r}{s^N - \sum_{k=0}^{N-1} a_k s^k} = \frac{\sum_{r=0}^{M} b_r s^{r-N}}{1 - \sum_{k=0}^{N-1} a_k s^{k-N}}$$

Signal flow graph:



(*N*>*M*)

State equation: $\boxed{\boldsymbol{\lambda}'(t) = \mathbf{A}\boldsymbol{\lambda}(t) + \mathbf{B}\mathbf{x}(t)}$

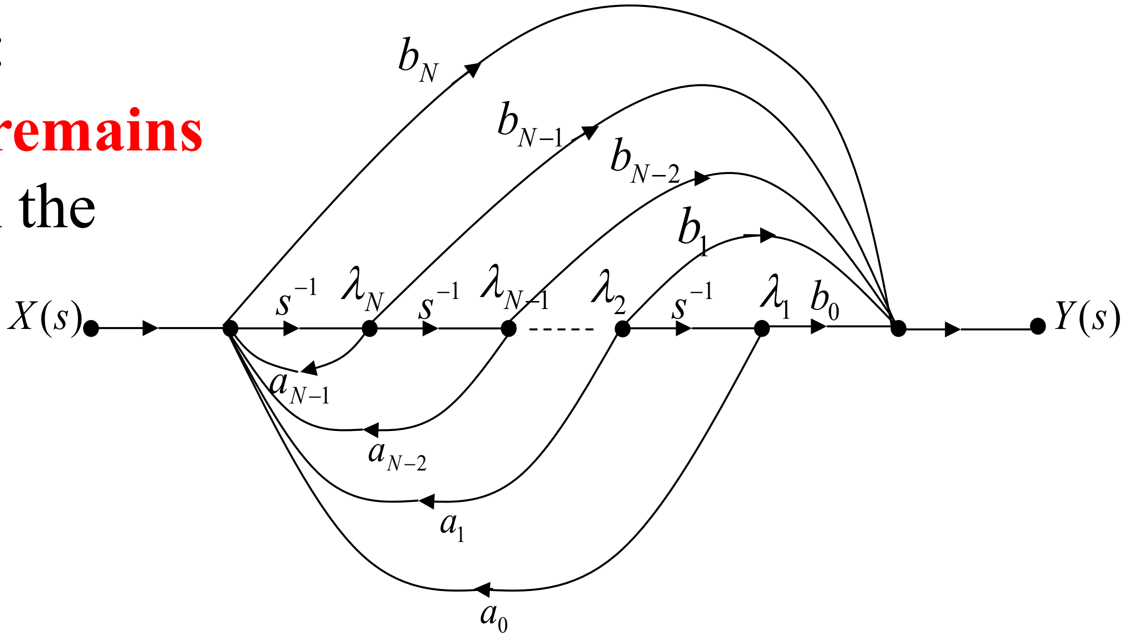
$$\underbrace{\begin{pmatrix} \dfrac{d\lambda_1(t)}{dt} \\ \dfrac{d\lambda_2(t)}{dt} \\ \vdots \\ \dfrac{d\lambda_{N-1}(t)}{dt} \\ \dfrac{d\lambda_N(t)}{dt} \end{pmatrix}}_{\boldsymbol{\lambda}'(t)} = \underbrace{\begin{pmatrix} 0 & 1 & 0 & \cdots & 0 & 0 \\ 0 & 0 & 1 & 0 & \cdots & 0 \\ \vdots & \cdots & & \cdots & & \vdots \\ 0 & \cdots & & \cdots & 0 & 1 \\ a_0 & a_1 & \cdots & \cdots & a_{N-2} & a_{N-1} \end{pmatrix}}_{\mathbf{A}} \underbrace{\begin{pmatrix} \lambda_1(t) \\ \lambda_2(t) \\ \vdots \\ \lambda_{N-1}(t) \\ \lambda_N(t) \end{pmatrix}}_{\boldsymbol{\lambda}(t)} + \underbrace{\begin{pmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ 1 \end{pmatrix}}_{\mathbf{B}} \left[x(t)\right]$$

Output Equation: $\boxed{\mathbf{y}(t) = \mathbf{C}\boldsymbol{\lambda}(t) + \mathbf{D}\mathbf{x}(t)}$

$\boxed{\mathbf{C} = [b_0 \quad b_1 \quad \cdots \quad b_{M-1} \quad b_M \quad 0 \quad \cdots \quad 0]}$

$\boxed{\textbf{Matrix } \mathbf{D} = \mathbf{0}}$

If ***N=M***, the signal flow graph of the system is:



At this point, **the state equation of the system remains the same as before**, but the **C** and **D** matrices in the output equation change:

$$y(t) = b_0\lambda_1(t) + b_1\lambda_2(t) + \ldots + b_{N-1}\lambda_N(t) + b_N \frac{d\lambda_N(t)}{dt}$$

$$\because \frac{d\lambda_N(t)}{dt} = a_0\lambda_1(t) + a_1\lambda_2(t) + \ldots + a_{N-1}\lambda_N(t) + x(t)$$

$$\therefore y(t) = (b_0 + b_N a_0)\lambda_1(t) + (b_1 + b_N a_1)\lambda_2(t) + \ldots + (b_{N-1} + b_N a_{N-1})\lambda_N(t) + b_N x(t)$$

Matrix **C** (at this point) $\mathbf{C} = \left[(b_0 + b_N a_0) \quad (b_1 + b_N a_1) \quad \ldots \quad (b_k + b_N a_k) \quad \ldots \quad (b_{N-1} + b_N a_{N-1})\right]$

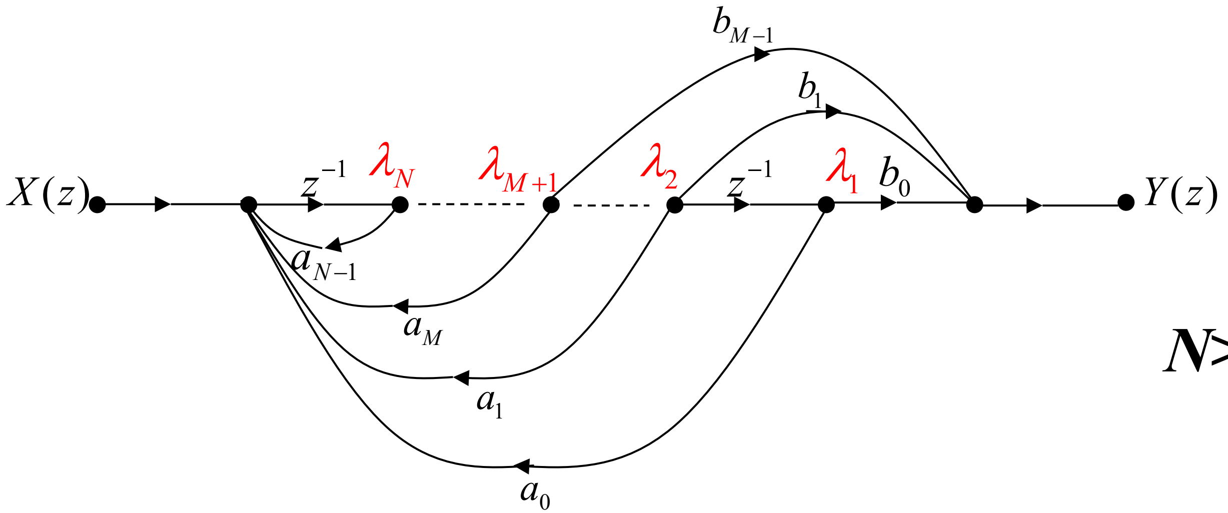
Matrix **D** (at this point) $\mathbf{D} = b_N$

Formulation of State Equations and Output Equations for Discrete-Time Systems

$$y(n)-\sum_{k=0}^{N-1} a_k y(n-(N-k))=\sum_{r=0}^{M} b_r x(n-(M-r))$$

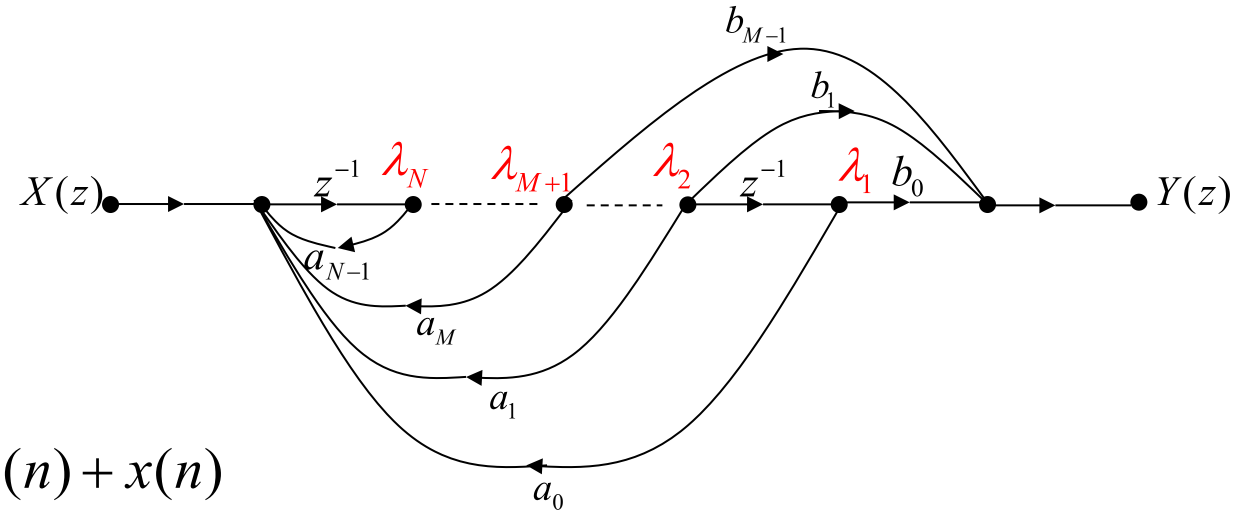
$$H(z)=\frac{\sum_{r=0}^{M} b_r z^{-(M-r)}}{1-\sum_{k=0}^{N-1} a_k z^{-(N-k)}}$$

(1) Draw the system's signal flow graph. In the graph, select **the output of the delay element** as the state variable (from the **output to input direction**)



N>M

(2) Formulate the State Equation



$$\lambda_1(n+1)=\lambda_2(n)$$

$$\lambda_2(n+1)=\lambda_3(n)$$

$$\vdots$$

$$\lambda_{N-1}(n+1)=\lambda_{N-2}(n)$$

$$\lambda_N(n+1)=a_0\lambda_1(n)+a_1\lambda_2(n)+\ldots+a_{N-1}\lambda_N(n)+x(n)$$

Matrix form:
$$\underbrace{\begin{pmatrix}\lambda_1(n+1)\\ \lambda_2(n+1)\\ \vdots\\ \lambda_{N-1}(n+1)\\ \lambda_N(n+1)\end{pmatrix}}_{\boldsymbol{\lambda}(n+1)}=\underbrace{\begin{pmatrix}0 & 1 & 0 & \cdots & 0 & 0\\ 0 & 0 & 1 & 0 & \cdots & 0\\ \vdots & \cdots & & \cdots & & \vdots\\ 0 & \cdots & & \cdots & 0 & 1\\ a_0 & a_1 & \cdots & \cdots & a_{N-2} & a_{N-1}\end{pmatrix}}_{\mathbf{A}}\underbrace{\begin{pmatrix}\lambda_1(n)\\ \lambda_2(n)\\ \vdots\\ \lambda_{N-1}(n)\\ \lambda_N(n)\end{pmatrix}}_{\boldsymbol{\lambda}(n)}+\underbrace{\begin{pmatrix}0\\ 0\\ \vdots\\ 0\\ 1\end{pmatrix}}_{\mathbf{B}}x(n)$$

State Equation: $\boxed{\boldsymbol{\lambda}(n+1)=\mathbf{A}\boldsymbol{\lambda}(n)+\mathbf{B}x(n)}$

Matrix A and B are identical to the coefficient matrices in the state equation of continuous-time systems.

Output Equation: $\boxed{y(n) = \mathbf{C}\boldsymbol{\lambda}(n) + \mathbf{D}x(n)}$

If ***N>M***,

$$y(n) = b_0\lambda_1(n) + b_1\lambda_2(n) + \ldots + b_{M-1}\lambda_M(n) + b_M\lambda_{M+1}(n)$$

$$\mathbf{C} = [b_0 \quad b_1 \quad \cdots \quad b_{M-1} \quad b_M \quad 0 \quad \cdots \quad 0]$$

D=0

If ***N=M***,

$$y(n) = (b_0 + b_N a_0)\lambda_1(n) + (b_1 + b_N a_1)\lambda_2(n) + \ldots$$
$$+ (b_{N-1} + b_N a_{N-1})\lambda_N(n) + b_N x(n)$$

$$\mathbf{C} = \left[(b_0 + b_N a_0) \quad (b_1 + b_N a_1) \quad \ldots \quad (b_k + b_N a_k) \quad \ldots \quad (b_{N-1} + b_N a_{N-1})\right]$$

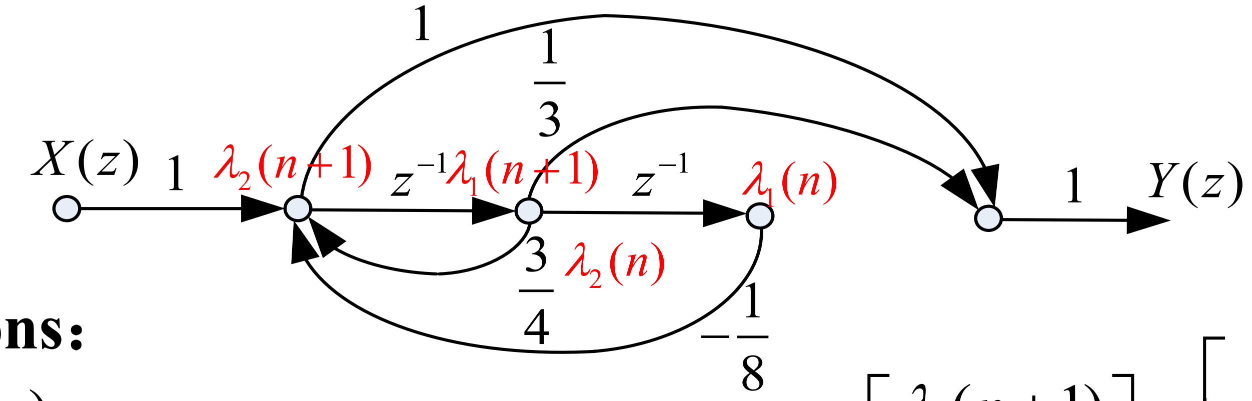
$$\mathbf{D} = b_N$$

Matrices C and D are identical to the coefficient matrices in the output equation of continuous-time systems.

Problem 8-1: Draw the signal flow graph of the system based on the following difference equation, and write the state equations and output equation in matrix form.

$$y(n)-\frac{3}{4}y(n-1)+\frac{1}{8}y(n-2)=x(n)+\frac{1}{3}x(n\text{-}1)$$

Solution:



state equations:

$$\begin{cases}\lambda_1(n+1)=\lambda_2(n)\\ \lambda_2(n+1)=-\frac{1}{8}\lambda_1(n)+\frac{3}{4}\lambda_2(n)+x(n)\end{cases}$$

$$\begin{bmatrix}\lambda_1(n+1)\\ \lambda_2(n+1)\end{bmatrix}=\underbrace{\begin{bmatrix}0 & 1\\ -\frac{1}{8} & \frac{3}{4}\end{bmatrix}}_{\mathbf{A}}\begin{bmatrix}\lambda_1(n)\\ \lambda_2(n)\end{bmatrix}+\underbrace{\begin{bmatrix}0\\ 1\end{bmatrix}}_{\mathbf{B}}x(n)$$

output equation:

$$y(n)=\underbrace{\begin{bmatrix}-\frac{1}{8} & \frac{13}{12}\end{bmatrix}}_{\mathbf{C}}\begin{bmatrix}\lambda_1(n)\\ \lambda_2(n)\end{bmatrix}+\underbrace{1}_{\mathbf{D}}\cdot x(n)$$